Sujith S

Research Analyst | Machine Learning · LLM Inference · Prompt Engineering · Python
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Summary

Computational Statistics and Data Analytics student at VIT University (2022–2027) with a peer-reviewed publication applying ML to financial time-series forecasting. Hands-on experience implementing classical ML models (classification, regression, time-series) and LLM-based inference pipelines using few-shot, chain-of-thought, and instruction-based prompting; skilled in offline evaluation, benchmarking, experiment documentation, and cross-functional collaboration to deliver scoped ML components end-to-end.

Education

Int.MSc Computational Statistics and Data Analytics VIT University | 2022–2027

Relevant Coursework: Machine Learning, Statistical Modeling, Data Structures & Algorithms, Database Systems, Financial Analytics, Distributed Systems, Computer Networks, Cloud Computing, Software Engineering

CS50 Programs Harvard CS50 (Online)

Completed CS50x, CS50P, CS50AI, CS50SQL, and CS50R covering Python, SQL, AI, algorithms, and full-stack development.

Skills

Languages: Python, SQL, R, Java, JavaScript, TypeScript

Machine Learning: Scikit-learn, XGBoost, PyTorch, TensorFlow, Classification, Regression, Time-Series Forecasting, Feature Engineering, Hyperparameter Tuning, Offline Evaluation & Benchmarking, MLflow, Statsmodels

Generative AI & LLMs: LangChain, LangGraph, OpenAI APIs, Hugging Face Transformers, Prompt Engineering (few-shot, chain-of-thought, instruction-based), RAG, FAISS, ChromaDB, AutoGen, CrewAI

Data Engineering & Analytics: Apache Kafka, Apache Spark, Apache Airflow, dbt, ETL/ELT, PostgreSQL, Redis, Delta Lake, Pandas, NumPy, Statistical Analysis, Correlation Analysis

Cloud & Infrastructure: AWS (Lambda, S3, RDS, EKS, EventBridge), GCP, Azure, Docker, FastAPI, REST APIs, CI/CD

Research & Publications

- **Stock Market Analysis and Prediction using Machine Learning** – El Ruha International Conference, 14^{th} Proceedings, pp. 730–740, 2025. Implemented classification and regression ML pipelines for financial time-series forecasting; designed feature engineering stages, benchmarked predictive models across multiple algorithms, and evaluated performance across real-world market regimes. [Code] [Proceedings]

Projects

Demand Forecasting System GitHub

Python, Scikit-learn, XGBoost, Prophet, Statsmodels, Pandas, NumPy, FastAPI, PostgreSQL, Redis, MLflow, Docker

- Implemented classical ML models (regression, ensemble, time-series) for sales and inventory demand prediction; engineered lag features, rolling averages, seasonal decomposition, and holiday regressors to improve predictive accuracy across product categories.
- Conducted offline benchmarking across 4 model families (Prophet, XGBoost, SARIMA, Ridge); deployed automated retraining pipelines with drift detection and confidence interval estimation; tracked all experiments and model performance metrics using MLflow for reproducibility.

Autonomous Research Agent GitHub

Python, LangChain, LangGraph, OpenAI APIs, Hugging Face Transformers, FAISS, ChromaDB, FastAPI, PostgreSQL, Redis, Docker

- Engineered LLM-based inference pipelines applying few-shot, chain-of-thought, and instruction-based prompting techniques to decompose multi-step research queries, execute multi-hop reasoning, and synthesize structured analytical reports; implemented RAG with FAISS vector search for grounded knowledge retrieval.

- Designed and ran offline evaluations benchmarking prompt strategies against ground-truth corpora; iteratively optimized prompts across few-shot, CoT, and zero-shot baselines; documented experiment results across prompt variants to support systematic model review and performance tracking.

Social Media Sentiment Lakehouse

GitHub

Python, Apache Kafka, Apache Spark Streaming, Delta Lake, Hugging Face Transformers, Scikit-learn, NLTK, MLflow, FastAPI, PostgreSQL, Docker

- Developed NLP classification pipelines using Hugging Face Transformers for real-time sentiment classification, emotion detection, and topic modeling at scale; applied statistical correlation analysis to identify engagement patterns; tracked and benchmarked ML model performance using MLflow across multiple classification approaches.

HR Workforce Intelligence Platform

GitHub

Python, Scikit-learn, XGBoost, Pandas, NumPy, FastAPI, PostgreSQL, Redis, MLflow, Docker, Power BI, Tableau

- Implemented ML-based classification models for employee attrition prediction and retention scoring; engineered workforce KPIs covering productivity, engagement, and utilization to support data-driven business decisions; evaluated and benchmarked models to identify optimal classification approach for operational deployment.

Work Experience

Founder

Shoot the Breeze | 2023 – Present

- Designed and deployed AI-powered applications integrating LLM inference and classical ML pipelines; owned full SDLC from requirements, system design, and implementation through evaluation, experiment documentation, and production monitoring; briefed technical progress to cross-functional stakeholders.

App/Web Developer

CropSky | VIT-TBI Research & Innovation Hub | 2024 – Present

- Developed scalable REST APIs and ML-integrated backend systems at a university-incubated research startup; collaborated cross-functionally with research scientists and engineers to translate business requirements into data-driven features; applied statistical analysis to convert research outputs into production-ready analytical modules.

Certifications & Achievements

- **Google Generative AI Leader**
- **Google Cloud Computing Foundations**
- **FreeCodeCamp Legacy Full Stack**
- **OCI 2025 Certified Developer Professional**
- AWS **Solutions Architect** and **Developer**
- **Software Engineer Certification - HackerRank**